

A novel multi-objective optimization of mass dampers for controlling the vortex-induced vibration in bridges

Wen-Peng Xu, Jun Dai^{*}, Zhao-Dong Xu^{*}

China-Pakistan Belt and Road Joint Laboratory on Smart Disaster Prevention of Major Infrastructures, Southeast University, Nanjing 210096, China
School of Civil Engineering, Southeast University, Nanjing 210096, China

ARTICLE INFO

Keywords:

Multi-objective optimization
Optimal design
Vortex-induced vibration
Mass damper

ABSTRACT

This paper investigates an optimization framework of the four mass dampers for controlling the vortex-induced vibration (VIV) in long-span bridges, including the tuned mass damper (TMD), the series double tuned mass dampers (DTMD) and the inerter-equipped tuned mass damper (ITMD). The equations of motion of the bridge-damper system subjected to the vortex-induced force are established. The vortex-induced amplitude of the girder is analytically given. Furthermore, the optimal design is implemented by taking the vortex-induced amplitude as the constraint condition. Introducing two optimization objectives, the stroke index and the robustness index, a novel bi-objective optimization framework with constraints is proposed and then solved by multiple objective particle swarm optimization (MOPSO). A comparative analysis of the four mass dampers with varying total mass ratios is performed. The results show that the vortex-induced amplitude of the girder is greatly reduced to satisfy the requirement after installing the four optimized mass dampers. Both performances of the two optimization objectives are improved with the increase of the total mass ratio. The inertial device parallel to the spring and dashpot provides no advantage in controlling the VIV in bridges. The DTMD has dramatic robustness to changes in the bridge natural frequency and cannot strongly amplify its stroke, especially under the condition of a small mass ratio. The framework provides an excellent balance between control effect and stability in the VIV of bridges for the optimal design of the mass dampers.

1. Introduction

Vortex-induced vibration (VIV) is a common form of vibration that occurs in long-span bridges at modest wind velocities [1–3]. Excessive vortex-induced amplitude will be generated when the vortex shedding frequency approaches the natural frequency of the bridge, which may drop the service life of the bridge and even cause structural damage [4,5]. Therefore, extensive research has been investigated on the mitigation of adverse vibration of bridges and various devices have been developed for vibration control [6–16].

The tuned mass damper (TMD) has been confirmed to be one of the simplest and most effective auxiliary devices for controlling structural vibration [17,18]. The TMD consists of a small oscillator attached to the structure (primary structure), which provides an additional free-to-vibrate mass to the dynamically excited primary structure to suppress its oscillatory motion [19]. A considerate parameter selection of the spring and damping is absolutely required to make the TMD more

effective [20]. Den Hartog [21] firstly derived the analytical solution of the optimal TMD parameters subjected to harmonic excitation by using the fixed-point theory. Warburton [22] analytically obtained the analytical solution of the optimal parameters for the TMD system under white noise force and acceleration excitation. The numerical searching method is normally adopted by researchers when it is difficult to get closed-form expressions for the TMD parameters [20,22]. A meta-heuristic algorithm was selected by Bekdas [23] to explore the optimal TMD parameters of the mass, stiffness and damping coefficient. Pourzeynali [24] proposed a robust multi-objective optimization design for the TMD to control structural vibrations against earthquakes. Jiménez-Alonso [25] used a multi-objective genetic algorithm for the robust optimization design of the TMD to mitigate pedestrian-induced vibrations of footbridges considering the uncertainties associated with their modal properties. Lavan [26] presented a multi-objective optimization formulation for the design of the TMD under a base excitation or an external load. The optimization aimed to simultaneously minimize the

^{*} Corresponding authors at: China-Pakistan Belt and Road Joint Laboratory on Smart Disaster Prevention of Major Infrastructures, Southeast University, Nanjing 210096, China.

E-mail addresses: daijun4242@163.com (J. Dai), zhdxu@163.com (Z.-D. Xu).

<https://doi.org/10.1016/j.engstruct.2023.115761>

Received 17 August 2022; Received in revised form 31 December 2022; Accepted 2 February 2023

Available online 17 February 2023

0141-0296/© 2023 Elsevier Ltd. All rights reserved.

structural response, the TMD mass and the TMD stroke.

The TMD so far has been successfully and widely applied in the field of wind-induced vibration control in bridges [27–29]. Gu et al. [30] investigated the buffeting mitigation of the Yangpu Bridge by using the TMD. Fujino and Yoshida [31] incorporated the TMD into a continuous steel box girder bridge to control the VIV. Chen and Kareem [32] studied the efficiency of the TMD for bridge flutter control and obtained the optimal parameters of the TMD. Domaneschi et al. [33] adopted the TMD to control the buffeting of suspension bridges. The results revealed that installing the TMD at the mid-span of the girder can decrease the bridge response.

However, the problems of the TMD detuning and narrow application frequency bandwidth urge researchers to pay attention to the multiple TMDs for higher vibration control performance and robustness [34,35]. The multiple TMDs can be classified into the series multiple TMDs and the parallel multiple TMDs [36]. Iwanami and Seto [37] optimized a parallel double TMDs according to the H_∞ criterion and the effectiveness of the damper in reducing the transmissibility at resonance was investigated. Xu and Igusa [38] proposed a parallel multiple TMDs with distributed natural frequencies. The results indicated that the parallel multiple TMDs has better effectiveness and robustness in reducing the structure vibration than the TMD. Nevertheless, an obvious shortcoming of applying the parallel multiple TMDs is the limitation of installation space [39]. Zuo [40] reported a series double TMDs (DTMD) to mitigate the vibration of the primary structure for random and harmonic vibration, and the study indicated that the DTMD is more effective and robust than the TMD, the viscoelastic TMD, and its parallel counterpart. Asami [36] optimized both the series and parallel double TMDs based on three design criteria (H_∞ , H_2 and stability maximization criterion). Ok et al. [41] investigated the optimal parameters of the bi-tuned mass dampers based on the two optimization objectives of the nominal and robust performance index. Furthermore, research revealed that series multiple TMDs are more effective and robust than parallel multiple TMDs [40]. The above critical literature reviews show that the multiple TMDs provide more effectiveness in suppressing vibration than the TMD. Despite that, to date, a few of the literature concerns the installation of the multiple TMDs in bridges for VIV control.

Recently, a novel device dubbed inerter drew the attention of researchers and was introduced to the vibration control of structures field due to its reliable high-performance and simple mechanism [42]. This mechanical device, firstly introduced by Smith [43], has the capacity to amplify its physical mass by converting linear motion into rotational motion [20]. Subsequently, researchers combined this inertial device with different conventional dampers to form various types of inerter-equipped dampers and optimized them to improve their performance [19,44–46]. Ikago et al. [44] proposed a tuned viscous mass damper (TVMD) and derived the closed-form expressions of optimal frequency and damping ratio by the fixed-point theory. Lazar et al. [45] adopted a tuned inerter damper (TID) to mitigate the seismic response of multi-story buildings which is consist of the inerter device, damping and stiffness elements. Marian and Giaralis [19] incorporated the inerter device with the TMD to form a tuned mass damper inerter (TMDI), in which the inerter device links the secondary mass to a different location from the location that the mass is suspended by the spring and dashpot. The TMDI was installed into an undamped structure subjected to white noise and harmonic excitations. The results indicated that the TMDI is more effective than the TMD in reducing the structure vibration. Pietrosanti et al. [46] further analyzed the effectiveness and robustness of different inerter-equipped dampers (namely TVMD, TMDI and TID) under earthquake base excitations. Taflanidis and Giaralis [47] developed a multi-objective design method for the inerter-equipped damper to determine the compromise between the objectives of vibrations mitigation and the inerter-equipped damper forces. The appropriateness of modelling assumptions for the design of the inerter-equipped dampers in earthquake engineering applications were further assessed. Patsialis [48] proposed a bi-objective inerter-equipped damper design

framework for linearly behaving buildings to balance structural performance and the inerter-equipped damper forces.

The inerter-equipped damper has been applied to bridges for wind-induced vibration control in some recent studies. Xu et al. [8] investigated the inerter-equipped TMD (ITMD) with the inertial device parallel to the spring and dashpot to mitigate the VIV of bridges. The results indicated that the ITMD can reduce the VIV responses of the bridge deck. And four different inerter-equipped dampers were further examined to control the VIV [49]. Dai et al. [13] proposed a Maxwell tuned mass-damper-inerter (M–TMDI), in which the dashpot element is replaced by the Maxwell element, to control the VIV in bridges, and derived the optimal parameters of the M–TMDI with any inerter location on the bridge deck.

This paper proposes a double tuned mass damper with inerter (DTMDI) to control the VIV in bridges, which is a combination of three mass dampers including the TMD, the DTMD and the ITMD. The equations of motion of the bridge-DTMDI system are established, and a new multi-objective optimization framework with constraints is proposed. The two optimization objectives of the stroke index and the robustness index for the four mass dampers are systematically studied and compared. This paper is arranged as follows: In Section 2, the equations of motion of the bridge-DTMDI system under the vortex-induced force are established; The optimization objectives are proposed and a novel constrained bi-objective optimization method is constructed in Section 3; The stroke index and the robustness index for the four mass dampers with the optimal parameters are discussed in Section 4; Section 5 summarizes the main conclusion.

2. Vortex-induced vibration characteristics of the bridge with DTMDI

2.1. Equations of motion of the bridge-DTMDI system

The bridge equipped with the DTMDI can be simplified into a representative three-DOF system, as shown in Fig. 1. DTMDI is composed of two TMDs in series denoted as TMD₁ and TMD₂. m_2 , k_2 , c_2 and m_3 , k_3 , c_3 are the mass, stiffness and damping of the TMD₁ and the TMD₂, respectively; y_1 is the displacement of the girder; y_2 denotes the relative displacement between the TMD₁ and the girder; y_3 denotes the relative displacement between the TMD₁ and the TMD₂; b is the inertance of the inerter device, which is connected between the TMD₁ and the girder. The resisting force of the inerter device can be calculated by:

$$F = b(\ddot{y}_u - \ddot{y}_d) \quad \# \quad (1)$$

where $\ddot{y}_u - \ddot{y}_d$ is the relative acceleration between the two terminals of the device.

The vortex-induced force F_{VIV} acting on the girder can be simulated by a semi-empirical nonlinear model [1] as:

$$F_{VIV} = \frac{1}{2} \rho U^2 (2B) \left[Y_1 \left(1 - e^{-\frac{Y_1^2}{B^2}} \right) \frac{\dot{y}_1}{U} + Y_2 \frac{y_1}{B} + \frac{1}{2} C_L \sin(\omega t + \theta) \right] \quad \# \quad (2)$$

where ρ is the air density; U is the mean wind velocity; B is the across-flow dimension of the section of the bridge; Y_1 , Y_2 , and ε are the aerodynamic parameters which characterize the negative damping ratio, aerodynamic stiffness, and self-limited amplitude, respectively; C_L denotes the lift coefficient resulting from the vortex-shedding effect; ω and θ are the vortex-shedding frequency and the initial phase.

This paper mainly focuses on the VIV in bridges in the lock-in region, since the large amplitude of the girder is always generated in this region [50]. And wind tunnel results show that the contributions of the aerodynamic stiffness and harmonic force on the vortex-induced amplitude could be neglected in the lock-in region [1]. Thus, the vortex-induced force can be simplified as:

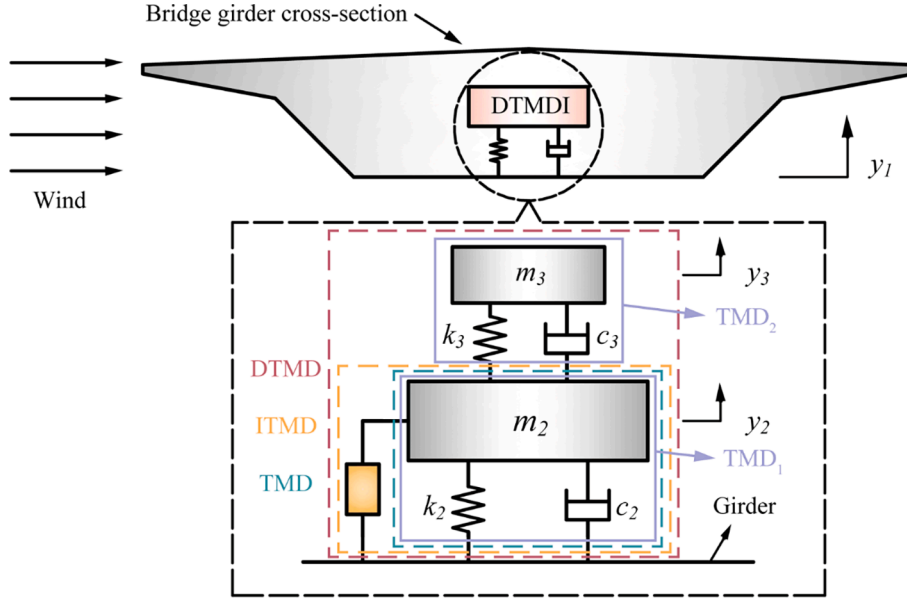


Fig. 1. The bridge-DTMDI system.

$$F_{VIV} = \rho U B Y_1 \left(1 - \varepsilon \frac{y_1^2}{B^2} \right) \dot{y}_1 \# \quad (3)$$

Vortex-induced resonance of bridges is a single-mode vibration phenomenon that usually occurs in one of the bridge modes [3]. To facilitate the analytical solution of the equations of motion, $y = \phi q$ is introduced. y is the displacement response of the bridge-DTMDI system; ϕ is the bridge mode shape in which the VIV occurs; q is the mode coordinate; The equations of motion of the bridge-DTMDI system subjected to the vortex-induced force can be written in the modal coordinate as:

$$\ddot{q}_1 + 2\xi_1 \omega_1 \dot{q}_1 + \omega_1^2 q_1 = \bar{Y}_1 (\alpha_2 - \bar{\varepsilon} \alpha_4 q_1^2) \dot{q}_1 - \mu_2 (\ddot{q}_1 + \ddot{q}_2) - \mu_3 (\ddot{q}_1 + \ddot{q}_2 + \ddot{q}_3) \# \quad (4)$$

$$\ddot{q}_1 + \ddot{q}_2 + 2\xi_2 \omega_2 \dot{q}_2 + \omega_2^2 q_2 + \frac{\beta}{\mu_2} \ddot{q}_2 = -\frac{\mu_3}{\mu_2} (\ddot{q}_1 + \ddot{q}_2 + \ddot{q}_3) \# \quad (5)$$

$$\ddot{q}_1 + \ddot{q}_2 + \ddot{q}_3 + 2\xi_3 \omega_3 \dot{q}_3 + \omega_3^2 q_3 = 0 \# \quad (6)$$

Where $\xi_i (i = 1, 2, 3)$ and $\omega_i (i = 1, 2, 3)$ are the damping ratios and frequencies of the bridge, the TMD₁ and TMD₂, respectively; $\mu_2 = M_2/M_1$ and $\mu_3 = M_3/M_1$ are the mass ratios of the TMD₁ and TMD₂ with respect to the bridge; $M_1 = \int_0^L m_1 \phi^2(x) dx$ is the modal mass for the bridge mode in which the VIV occurs; $M_2 = \int_0^L m_2 \phi^2(x) dx$ and $M_3 = \int_0^L m_3 \phi^2(x) dx$ are the effective mass of the TMD₁ and TMD₂; L is the span length of the bridge; $\bar{Y}_1 = \rho U B Y_1 / M_1$, $\alpha_2 = \int_0^L \phi^2(x) dx$, $\alpha_4 = \int_0^L \phi^4(x) dx$, $\bar{\varepsilon} = \varepsilon / B^2$ and $\beta = b / M_1$.

2.2. Analytical solution of the bridge-DTMDI system

The Krylov-Bogoliubov method [51] is applied to the nonlinear equations to obtain the vortex-induced amplitude of the girder. For the purpose of brevity, the detailed derivation process is provided in Appendix A. The vortex-induced amplitude A_1 and phase φ can be expressed as:

$$A_1 = \sqrt{\frac{M}{N} \left(1 - \frac{\frac{M}{N} - A_0^2}{A_0^2 e^{M_1} + \frac{M}{N} - A_0^2} \right)} \# \quad (7)$$

$$\varphi = -\frac{1}{2} \omega t \left[1 - \frac{\omega_1^2}{\omega^2} + \mu_2 (1 + P_{21}) + \mu_3 (1 + P_{21} + P_{31}) \right] + \varphi_0 \# \quad (8)$$

Where $M = \bar{Y}_1 \alpha_2 + \mu_2 \omega Q_{21} + \mu_3 \omega (Q_{21} + Q_{31}) - 2\xi_1 \omega_1$, $N = \bar{Y}_1 \bar{\varepsilon} \alpha_4 / 4$; Q_{21} , Q_{31} , P_{21} and P_{31} are given in Appendix A; A_0 and φ_0 are the initial displacement and the initial phase. The vortex-induced phase φ is assumed to be a slowly varying function of time. Thus, the derivative of φ is 0, namely $\dot{\varphi} = 0$. According to Eq. (8), the excitation frequency ω of the VIV can be obtained. It is noteworthy that multiple solutions will be generated, and the determination method is given in [4].

According to Eq. (7), when $t \rightarrow \infty$, $\lim_{t \rightarrow \infty} A_1 = \sqrt{\frac{M}{N}} = 2\sqrt{\frac{\alpha_2}{\bar{\varepsilon} \alpha_4}} \sqrt{1 - \frac{2\omega_1 (\xi_1 + \xi_{eq})}{\alpha_2 Y_1}}$, where $\xi_{eq} = -\frac{\omega}{\omega_1} [\mu_2 Q_{21} + \mu_3 (Q_{21} + Q_{31})]$, which is called the equivalent damping ratio in the VIV mitigation by Dai [3]. When $\xi_{eq} < \frac{\alpha_2 \bar{Y}_1}{2\omega_1} - \xi_1$, $\lim_{t \rightarrow \infty} A_1$ is a positive real number, which indicates that the vortex-induced amplitude of the girder will tend to a stable value. And a limit cycle oscillation (LOC) with a steady-state amplitude will be generated, whose amplitude varies with the damper parameters. When $\xi_{eq} \geq \frac{\alpha_2 \bar{Y}_1}{2\omega_1} - \xi_1$, the value of $\lim_{t \rightarrow \infty} A_1$ is 0 or imaginary. Theoretically, the vortex-induced amplitude of the girder will decrease gradually until it disappears. However, with the decline of the amplitude, the third term C_L in Eq. (2) cannot be ignored anymore. When the vortex-induced force is dominated by the third term, the girder is more inclined to be excited by a harmonic excitation, which also means that the girder will continue to vibrate with a small amplitude. Thus, maximizing the equivalent damping is not the best method to develop the potential of dampers. An expanded novel concept is proposed to search for a more comprehensive damper performance, which is more suitable for practical applications.

2.3. Bridge model

To further evaluate the characteristics of the bridge-DTMDI system, a bridge adopted from [13] is adopted as a case. The model is the deep-water non-navigable bridge of the Hong Kong-Zhuhai-Macao Bridge, a 6×110 m continuous steel box-girder bridge, with a height of 4.5 m and a width of 33.1 m section. The generalized mass and natural frequency for the fundamental vertical mode are 8888.9 tons and 0.806 Hz. The damping ratio of the bridge is assumed as 0.3 %. The key structural properties of the bridge and the values of the aerodynamic parameters

are summarized in Table 1.

Wind tunnel test results of the scaled sectional and full-bridge aeroelastic models showed that a vertical vortex-induced resonance of the bridge with a maximum amplitude of 150 mm–160 mm will occur at the wind velocity of 27 m/s–35 m/s according to [52]. And the amplitude of the girder far exceeds the requirement of the Chinese code (50 mm) [53]. The aerodynamic parameters are identified by utilizing a simplified decay-to-resonance method. The vortex-induced resonance of the bridge can be simulated by solving fundamental structural dynamic equations. The Newmark- β method in conjunction with the direct iteration method [54] is applied as the numerical method in simulation. The initial amplitude of the girder is assumed as 50 mm. Fig. 2 shows the simulated vortex-induced resonance of the bridge. It can be seen that the vortex-induced amplitude gradually exhibits an oscillation with a steady-state amplitude of 150 mm, which indicates that the semi-empirical nonlinear model can capture the limit cycle phenomenon of the VIV.

3. Optimal design of dampers

3.1. Optimization objective function

The displacement response of the bridge is the concerned demand parameter in the bridge-damper system, which is most related to structural failure. In the present study, the amplitude of the girder serves as a significant indicator of the bridge performance for optimization. Moreover, the stroke of the mass block in the mass damper cannot be neglected because an excessive stroke of the mass block can drop the service life of dampers, and may even lead to the collision between the mass blocks and the bridge, resulting in structural damage. The robustness of the bridge-damper system is another crucial index for evaluating the performance of dampers. It is difficult to guarantee that the working conditions in practical application are perfectly consistent with the theoretical model because some uncertainties unavoidably exist in bridge engineering. Apparently, these uncertainties will affect the overall performance of bridges. Based on the above, three goals of the damper utilized in the VIV control of bridges and corresponding optimization objective functions are proposed in this section.

The main goals of the damper are as follows:

- (1) Reduce the vortex-induced amplitude of the girder to satisfy the requirement;
- (2) Minimize the stroke of the mass block in the mass damper when the VIV occurs in the bridge;
- (3) Enhance the stability of the control effect with the uncertain structural properties, namely the reinforcement of the system robustness.

Three dimensionless optimization objective functions for the bridge-DTMDI system are formulated simultaneously to characterize the goals.

(i) The amplitude index f_A

The amplitude index f_A is defined by Eq. (9) as follows, which is a high-priority to focus on in the parameter optimization of dampers.

$$f_A = \frac{A_1}{A_{limit}} = \frac{2}{A_{limit}} \sqrt{\frac{\alpha_2}{\bar{\epsilon}\alpha_4}} \sqrt{1 - \frac{2\omega_1 \left(\xi_1 - \frac{\omega}{\omega_1} [\mu_2 Q_{21} + \mu_3 (Q_{21} + Q_{31})] \right)}{\alpha_2 \bar{Y}_1}} \# \quad (9)$$

Where A_1 is the vortex-induced amplitude of the girder in the VIV, which can be derived by Eq. (7); A_{limit} is the upper limit of the

requirement (namely 50 mm in this paper).

(ii) The stroke index f_S

The definition of the stroke index f_S can be expressed by Eq. (10):

$$f_S = \frac{A_2 + A_3}{B} = \frac{A_1}{B} \left(\sqrt{P_{21}^2 + Q_{21}^2} + \sqrt{P_{31}^2 + Q_{31}^2} \right) \# \quad (10)$$

Where A_2 and A_3 are the stroke of the TMD₁ and TMD₂, respectively, which can be obtained according to Eqs. (A.6) and (A.8); B is the across-flow dimension of the section of the bridge.

Referring to the stroke of the mass block as the stroke of the mass damper, it is obvious that the stroke index f_S is proportional to the stroke of the DTMDI.

(iii) The robustness index f_R

In order to assess the robustness of the bridge-DTMDI system, a quantitative formulation should be addressed. Deghan Niri et al. [55] combined the flatness of the curve with the maximum dynamic amplification factor (DAF) as a quantitative robustness index of the TMD control effect. Dai et al. [3] selected the flatness of the equivalent damping ratio of the bridge as the index of system robustness and the probability of error occurrence is considered. In the present study, the vortex-induced amplitude of the girder is taken as the reference index of the system robustness, because the amplitude is a more intuitive performance parameter compared to others. The improved robustness index can be expressed by the following formula:

$$f_R = \int_{e_1}^{e_r} P(e) \tan^{-1} \left(\left| \frac{A_1 - A_{1,e}}{e} \right| \right) de \# \quad (11)$$

Where A_1 and $A_{1,e}$ are the vortex-induced amplitude of the girder without and with the relative parameter error e , which refer to the error of the bridge natural frequency; $P(e)$ denotes the probability distribution density of the relative parameter error.

Fig. 3 shows the relationship between the vortex-induced amplitude of the girder and the bridge natural frequency. Introducing $\theta(e) = \tan^{-1} |(A_1 - A_{1,e})/e|$, obviously, the smaller value of $\theta(e)$ indicates a flatter curve, showing that the amplitude of the girder is more insensitive to the bridge natural frequency. Each optimization will make the minimum amplitude appear at the bridge initial natural frequency for contrast, namely point E in Fig. 3. As the bridge natural frequency deviates from the initial value ω_1 , the amplitude of the girder increases rapidly. The curve has two intersections M and N with line $y = A_{limit}$ (A_{limit} is the upper limit of the requirement). The horizontal distance from point M to point E is denoted as e_1 , while the corresponding distance from point N to point E is e_2 . Comparing the value of e_1 and e_2 , the smaller value is chosen as the effective length, namely e_L in Fig. 3. Finally, the robustness index f_R is calculated by integrating Eq. (11) on $[\omega_1 - e_L, \omega_1 + e_L]$. Assuming that the bridge natural frequency obeys a Gaussian distribution, and the coefficient of variation δ means the level of uncertainty of the natural frequency, which is taken as 5 %.

3.2. Constrained bi-objective optimization method (CBOM)

It is arduous to completely eliminate the VIV in bridges as mentioned in the previous section. In this sense, the priority is to optimize the damper parameters under the premise of ensuring that the amplitude of the bridge girder satisfies the requirement. Therefore, optimization objective function (i) can be optimized as a constraint instead of a goal, which can reduce the dimension of the optimization objectives and the computational efforts. The subsequent optimization design will be meaningless unless the amplitude of the girder meets the requirement, namely $f_A \leq 1$. Thus, the above optimization problem can be presented by the following constrained bi-objective optimization model:

$$\begin{aligned} \min F(x) &= \{f_S(x), f_R(x)\} \\ \text{s.t. } f_A(x) &\leq 1 \\ x &= (\mu_k, \lambda_2, \lambda_3, \xi_2, \xi_3, \beta) \in X \# \end{aligned} \quad (12)$$

Table 1

Structural and aerodynamic parameters.

B	ω_1	ξ_1	M_1	Y_1	ϵ
4.5 m	5.0624 rad/s	0.3 %	8889000 kg	15.674	2900

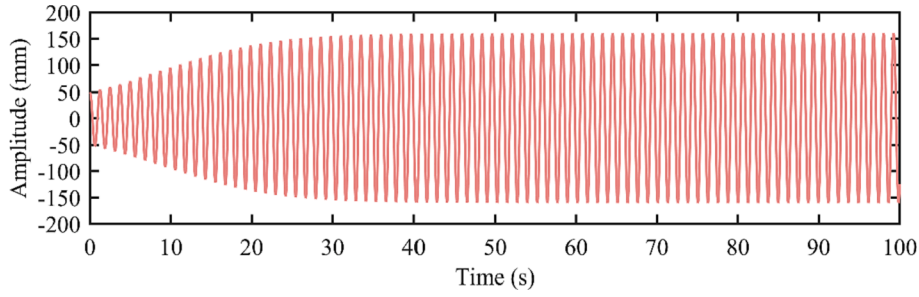


Fig. 2. Vortex-induced amplitude of the girder.

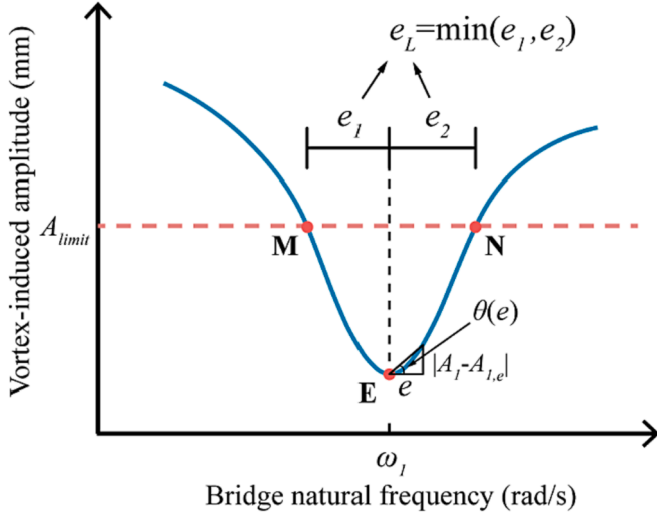


Fig. 3. Vortex-induced amplitude with respect to the bridge natural frequency.

Where $f_S(x)$, $f_R(x)$ and $f_A(x)$ denote the stroke index, the robustness index and the amplitude index, respectively; x and X denote the optimization variable and the variable space; μ_k is the mass ratio of the TMD₂ to the TMD₁; λ_2 and λ_3 are the frequency tuning ratios of the TMD₂ to the bridge and the TMD₁ to the bridge. ξ_2 and ξ_3 are the damping ratio of the TMD₁ and TMD₂, respectively, and β is the ratio of the inertial mass with respect to the modal mass of the bridge.

It is noticeable that the optimal value of ξ_2 in the DTMD is 0 in the reference [30], which is obtained when the primary structure is excited by a harmonic force and H_∞ optimization is adopted. For the VIV in bridges, the vortex-induced force on the girder is nonlinear, and the optimization objective is not the minimization of the DAF. Therefore, the value of ξ_2 will still be considered in this paper. Note that the above optimization design allows for the explicit treatment of the TMD, DTMD and ITMD as special cases of the DTMDI by setting $\mu_k = 0$, $\lambda_3 = 0$, $\xi_3 = 0$, $\beta = 0$ or $\beta = 0$ or $\mu_k = 0$, $\lambda_3 = 0$, $\xi_3 = 0$, respectively.

The solutions of the multi-objective optimization problem stated above, in principle, will be expressed as Pareto fronts, which are curves or surfaces plotted based on the objective function values of Pareto optimal individuals. And every point in the Pareto front is optimal in the sense that improvement in one objective function leads to deterioration in at least one of the remaining objective functions. So far, numerous multi-objective algorithms [56–60] have been proposed, such as fast and elitist multiobjective genetic algorithm [57] and multi-objective uniform-diversity genetic algorithm [58], to obtain a clear presentation of the Pareto fronts. Due to the excellent convergence and strong scalability compared with others, multiple objective particle swarm optimization (MOPSO) [61] is adopted to solve the aforementioned optimization problem. Besides, a penalty function is added to deal with the optimization problem with constraints. The constrained bi-objective

optimization method (CBOM) is depicted by the framework shown in Fig. 4.

4. Analysis of optimization results

In general, the mass of the mass damper affects the control performance. Therefore, μ is defined as the total mass ratio of the DTMDI, namely $\mu = \mu_2 + \mu_3$. The proposed optimization method is repeated while varying μ . In this paper, four different total mass ratios, 0.005, 0.01, 0.015, 0.02 are considered to cover the entire range of practical applications in bridges.

Fig. 5 displays the optimization results of the TMD, ITMD, DTMD and DTMDI in the two objective function spaces for the four total mass ratios considered, including 12 groups of different Pareto fronts. Each group of fronts is composed of 30 Pareto optimal solutions, for the sake of presentation, which is evenly distributed as far as possible in the spaces. Obviously, the stroke index f_S is inversely proportional to the robustness index f_R .

The Pareto fronts of the TMD are always distributed in the left middle region of Fig. 5 for each μ , representing the small stroke index f_S and the moderate robustness index f_R . In other words, the optimized TMD controls the amplitude of the girder with a small stroke and possesses a medium robustness to changes in the bridge natural frequency in comparison with the others. Note that in Fig. 5, the projection of the Pareto fronts for the TMD on the x-axis decreases with the increase of μ , which indicates that a smaller stroke of the TMD is enough to restrict the amplitude within the requirement by using a larger μ . This is consistent with the principle of momentum conservation. Besides, the increment of μ leads to an overall reduction of f_R for TMD.

The general trends of f_S and f_R against μ for the ITMD are similar to those in the TMD. It is noteworthy that the Pareto fronts of the ITMD are almost distributed directly above those of the TMD. That means the optimized TMD and ITMD have basically identical stroke behavior for a particular μ , whereas the stability to changes in the bridge natural frequency of the TMD is stronger.

For the DTMD, the Pareto fronts are distributed at the bottom in each subfigure of Fig. 5, namely the smallest robustness index f_R , which shows that the optimized DTMD has the most stable performance to changes of the bridge natural frequency. Note that for a particular μ , the projection of the Pareto fronts for the DTMD on the x-axis is far more widespread than that of the Pareto fronts for the TMD and the ITMD, since the DTMD contains two mass blocks resulting in greater stroke compared to the others.

The distribution of the Pareto fronts for the DTMDI are close to that of the DTMD. That means the optimized DTMDI and DTMD have similar stroke behavior and stability to changes in the bridge natural frequency. As for f_S and f_R of the DTMDI and DTMD, with the respect to μ , they have similar trends to that of the TMD and the ITMD.

4.1. Stroke analysis

To further investigate the stroke of the four mass dampers, Fig. 6

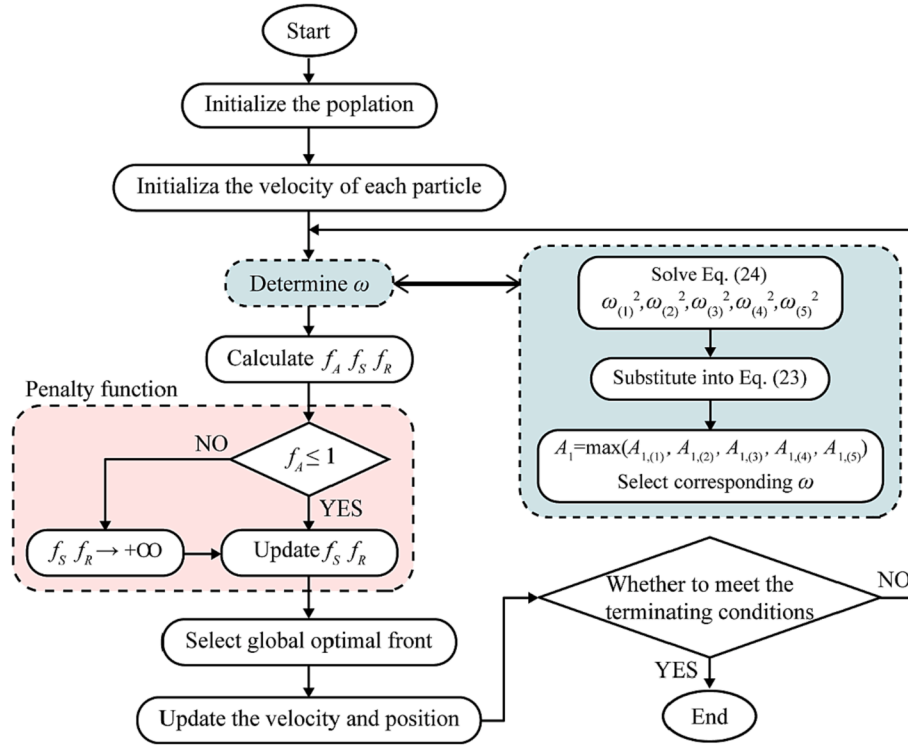


Fig. 4. Framework of the optimization method.

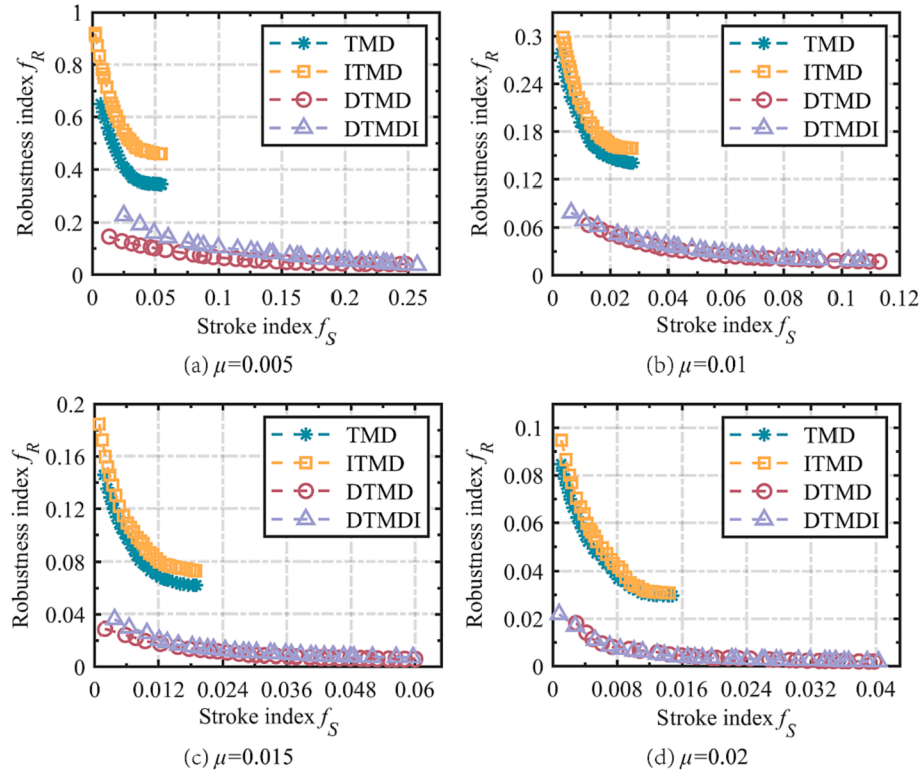


Fig. 5. Pareto fronts for the TMD, ITMD, DTMD and DTMDI under different total mass ratios.

presents the stroke index f_S for the TMD, ITMD, DTMD and DTMDI with respect to the total mass ratio μ . The length of the vertical line denotes the range of f_S and the center point represents the average value of f_S for each Pareto front. As shown, the range and average value of f_S for the four mass dampers are negatively correlated with μ . It is worth noting

that for each μ , the ranges of f_S for the four mass dampers overlap partially. That means the Pareto solutions in the overlapping area can make the strokes of the four mass dampers equal under the required amplitudes of the girder.

In reality, according to Eqs. (A.4), (A.6) and (A.8), the relationships

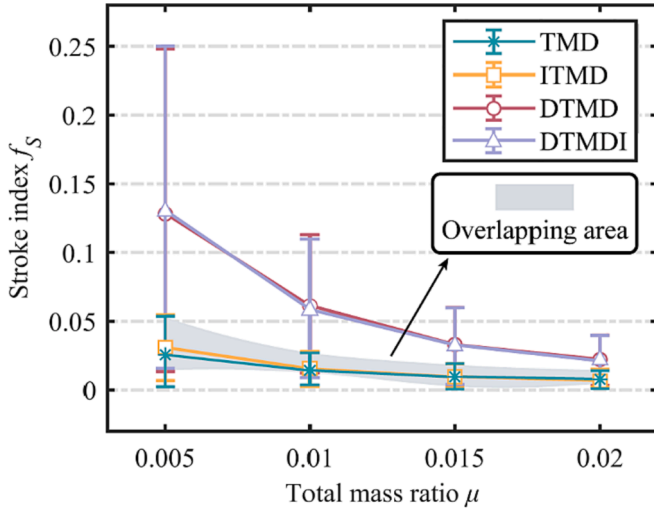


Fig. 6. Range and average value of f_s for the TMD, ITMD, DTMD and DTMDI.

between the amplitudes of the girder and the strokes of the four mass dampers for each Pareto solution are further given, as shown in Fig. 7. It can be seen that for the four mass dampers, the amplitude of the girder is almost linear with the stroke of the mass damper. The slope of the line for the DTMD and the DTMDI is higher than that of the line for the TMD and the ITMD, which illustrates that the stroke of the DTMD and the DTMDI is larger than that of the TMD and the ITMD under the same control effectiveness (namely the same amplitude of the girder). It should be noted that the stroke of the DTMD can be controlled to a small extent when its parameters are selected from the individuals of Pareto fronts with the smallest amplitudes of girder in Fig. 7(b). Furthermore, there exists a mapping between the points shown in Fig. 5 and Fig. 7. Indeed, each point of the Pareto fronts in Fig. 5 corresponds to a point

given in Fig. 7.

For fair comparison and intuitive observation, the individuals of Pareto fronts with corresponding amplitude of the girder close to 10 mm are selected as representative points, as displayed by the highlighted points in Fig. 7. The parameters of representative points are substituted into the dynamic equation Eqs. (4), (5), (6) and solved by the Newmark- β method in conjunction with the direct iteration method. Fig. 8 provides the displacement time histories of the girder equipped with the TMD, ITMD, DTMD and DTMDI when $\mu = 0.01$, respectively. A considerable control effect can be realized for the four mass dampers in comparison with the uncontrolled bridge displayed in Fig. 2. In addition, the values of the stable strokes in Fig. 8 are equal to the vertical co-ordinates of the representative points in Fig. 7(b), further verifying the correctness of the analytical solution in Section 2.

4.2. Robustness analysis

Fig. 9 shows the influences of the total mass ratio μ on the range and average value of the robustness index f_R for the TMD, ITMD, DTMD and DTMDI. Similar to the trend of f_s , the range and average value of f_R simultaneously decrease with the increment of μ . It demonstrates that the performance of the four mass dampers to changes in the bridge natural frequency tends to be more stable when μ is relatively large. Obviously, f_R for the DTMD and the DTMDI are significantly smaller than that for the TMD and the ITMD. Although the four f_R are close when μ is large, the entire Pareto solutions of the DTMD and the DTMDI still dominate the others in terms of f_R , as shown in Fig. 9, which indicates the DTMD and the DTMDI possess the stability that the TMD and the ITMD cannot achieve.

The parameters of the highlighted points in Fig. 7 are still selected to further comprehend the robustness of the bridge-DTMDI system. Introducing frequency ratio $\lambda = \omega_e/\omega_1$, ω_e denotes the bridge natural frequency with the relative parameter error e . Fig. 10 provides the variation of the amplitude of the girder equipped with the TMD, ITMD,

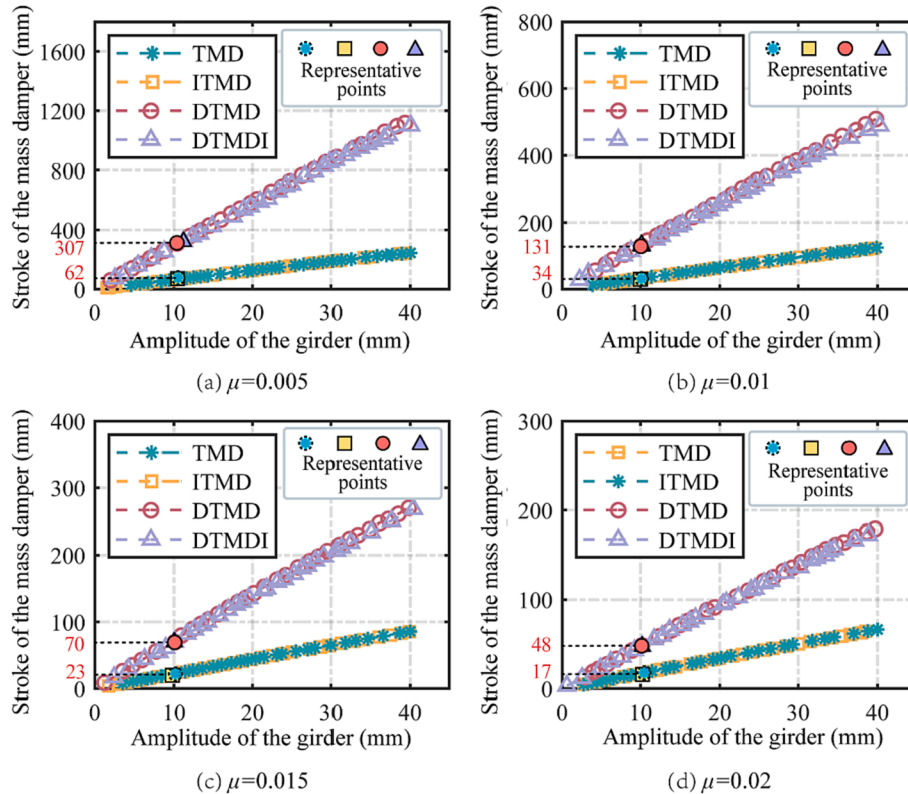


Fig. 7. Variation of the stroke of the mass damper with respect to the amplitude of the girder.

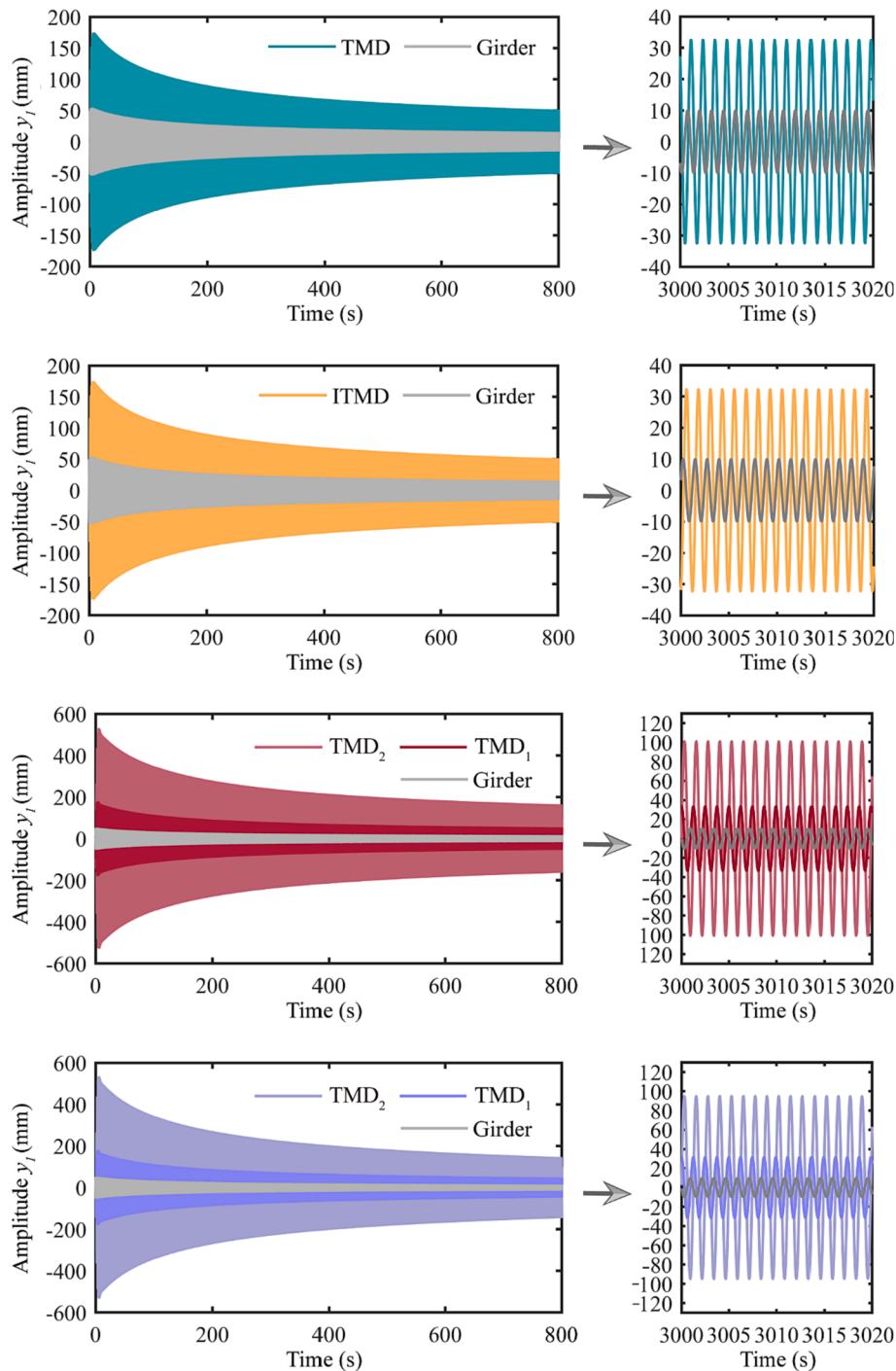


Fig. 8. Displacement time histories of the girder equipped with the TMD, ITMD, DTMD and DTMDI.

DTMD and DTMDI against the frequency ratio. It can be clearly seen that the curve shapes for the four mass dampers are similar. The minimum amplitudes of the girder equipped with the four mass dampers appear at $\lambda = 1$ with a basically identical value (about 10 mm), and the amplitude increases as λ moves away from 1. Apparently, the curve for the DTMD and the DTMDI is wider and flatter than that for the TMD and the ITMD. The range of λ where the vortex-induced amplitude of the girder is less than the requirement (50 mm) is defined as the applicable range. Note that the applicable range for the DTMD and the DTMDI is considerably larger than the corresponding range for the TMD and the ITMD. Table 2 tabulates the optimal parameters and the applicable range for the four mass dampers in Fig. 10. The applicable range for the DTMD is improved

with 68 %–168.8 %, 80.6 %–235.7 % and 5.3 %–28.8 % with respect to the TMD, the ITMD and the DTMDI. Interestingly, the DTMD yields the greatest enhancement when μ is smallest in terms of the applicable range.

4.3. Discussion for the optimization design of the mass damper

The above results show that the TMD is a competent choice, with its smaller stroke index f_s and moderate robustness index f_R , for controlling the VIV in bridges. For the ITMD, it is less robust than the TMD and the DTMD in terms of changes in the bridge natural frequency, even if its control effectiveness and stability have been demonstrated in some

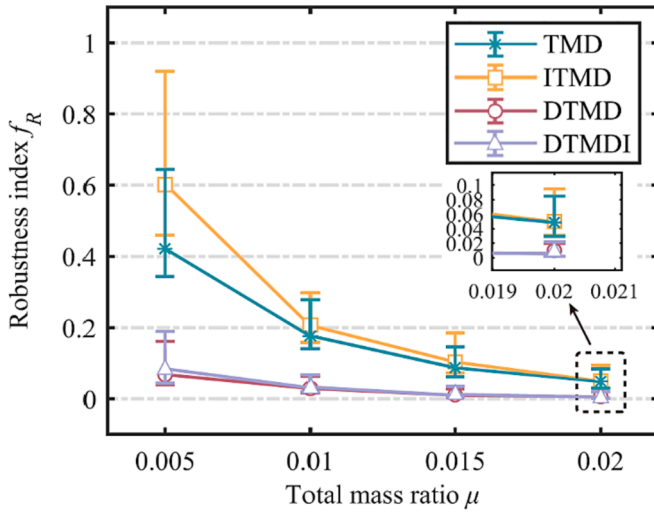


Fig. 9. Range and average value of f_R for the TMD, ITMD, DTMD and DTMDI.

literature [7,8]. In fact, it can be found in Table 2 that parameter β of the ITMD and the DTMDI is taken as the minimum value of 0.001 in the given range. The optimal parameters of the ITMD and the DTMDI are approaching that of the TMD and the DTMD, which means the TMD and the DTMD are the ideal situations for the ITMD and the DTMDI under the proposed optimization method. Therefore, the inertial device parallel to the spring and dashpot provides no advantage in controlling the VIV in bridges, which is also confirmed in [62,63]. The DTMD easily generates large strokes, which is restricted by girder height or installation space in box bridges. However, it possesses, with reasonable parameters, not only a small stroke (even close to the TMD), but also a dramatic stability to changes in the bridge natural frequency. It is an option worth considering.

Fig. 11 gives a comparison between the mass dampers optimized by

other optimization methods and the constrained bi-objective optimization method (CBOM) in this paper when $\mu = 0.01$, in terms of variation of the amplitude of the girder against the frequency ratio. It is worth noting that the parameters selected for those optimization methods are from the individuals of Pareto fronts with the smallest amplitudes of the girder in Fig. 7(b). A comparison of the TMD optimized by the Den Hartog formula and the CBOM is given in Fig. 11(a). It can be seen that for the original bridge natural frequency, namely $\lambda = 1$, the TMD optimized by both optimization methods effectively control the VIV in the bridge. When λ is in a small range close to 1, the Den Hartog formula has a modest advantage in terms of the amplitude of the girder, but the CBOM also provides satisfactory performance. However, as λ moves away from 1, the vertical coordinate of the curve for the Den Hartog formula increases rapidly. The control efficiency of the TMD optimized by the CBOM is significantly better when $\lambda < 0.942$ or $\lambda > 1.077$. For the entire range of λ , the amplitude of the girder equipped with the CBOM-optimized TMD is more stable and has a smaller value. Fig. 11(b) compares the three different DTMD optimization methods in [36] with the CBOM. The DTMD optimized by all optimization methods supplies excellent control for the VIV when $\lambda = 1$. The control efficiency of the CBOM-optimized DTMD outperforms the other three conventional optimization methods when $\lambda < 0.929$ or $\lambda > 1.089$. Considering the whole range of λ , it is clear that the CBOM is significantly superior to the other optimization methods in every aspect. In summary, compared to the conventional optimization method, the CBOM provides a better balance between control effect and stability in the VIV of the bridge for the optimal design of the mass dampers.

5. Conclusions

In this paper, a constrained bi-objective optimization method (CBOM) is proposed to optimize the parameters of four mass dampers controlling the VIV in the bridge. The analytical solution of the bridge-DTMDI system under vortex-induced force is derived. The optimization is solved by MOPSO and two optimization objectives of the four mass

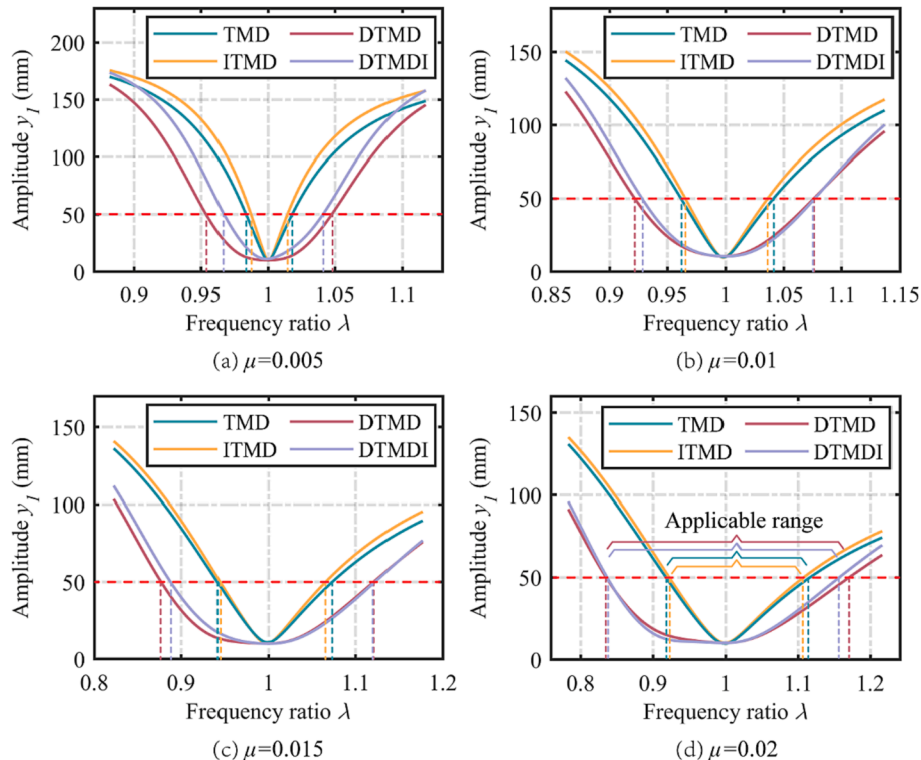
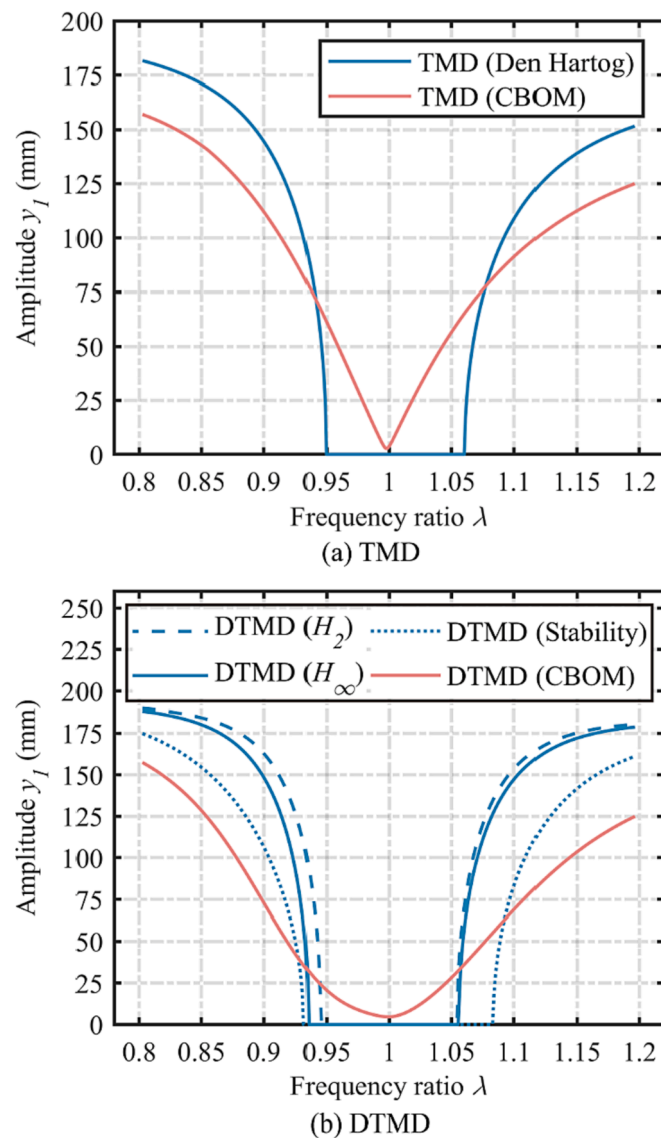


Fig. 10. Amplitude of the girder with respect to the frequency ratio.

Table 2

Pareto optimal parameters for the four mass dampers and the corresponding applicable ranges.

μ		μ_k	λ_2	λ_3	ξ_2	ξ_3	β	Applicable ranges
0.005	TMD		0.9922		0.0777			0.983–1.018
	ITMD		1.0892		0.071		0.001	0.987–1.015
	DTMD	0.039	1.0177	0.9791	0.0098	0.1405		0.954–1.048
0.01	DTMDI	0.0424	1.1249	0.9868	0.0016	0.1351	0.001	0.968–1.041
	TMD		0.9697		0.1546			0.96–1.041
	ITMD		1.0217		0.1474		0.001	0.965–1.037
	DTMD	0.0436	1.0043	0.9582	0.0922	0.1689		0.921–1.075
0.015	DTMDI	0.0541	1.0698	0.9611	0.07639	0.1719	0.001	0.93–1.075
	TMD		0.936		0.2309			0.941–1.074
	ITMD		0.9731		0.2234		0.001	0.945–1.065
	DTMD	0.0826	1.0012	0.9198	0.1449	0.2279		0.877–1.121
0.02	DTMDI	0.1036	1.0623	0.9208	0.1289	0.2409	0.001	0.889–1.121
	TMD		0.8878		0.3043			0.917–1.117
	ITMD		0.92		0.2977		0.001	0.922–1.108
	DTMD	0.1417	0.9998	0.868	0.1957	0.2957		0.836–1.172
	DTMDI	0.1383	1.0305	0.8726	0.1875	0.279	0.001	0.838–1.157

**Fig. 11.** Comparison of the amplitude of the girder with respect to the frequency ratio under different optimization methods.

dampers, termed as stroke index and robustness index, are compared. The following main conclusions are obtained:

(1) The vortex-induced amplitudes of the girder significantly decline to meet the requirement after equipping four mass dampers optimized by the CBOM.

(2) The stroke index is inversely proportional to the robustness index. With the increase of the total mass ratio, the performance of both optimization objectives for the four mass dampers is improved.

(3) For each total mass ratio, the ranges of stroke index for the four mass dampers overlap partially, which indicates that the Pareto solutions in the overlapping area can make the strokes of the four mass dampers equal under the required amplitudes of the girder.

(4) The robustness to changes in the bridge natural frequency for the DTMD is substantially higher than that for the TMD and the ITMD. The improvement effect is more remarkable with a smaller total mass ratio.

(5) The CBOM provides a better balance between control effect and stability in the VIV of bridges for the optimal design of the mass dampers compared to the conventional optimization method.

In addition, the robustness against changing bridge natural frequency for the DTMD can be dramatically improved without substantially increasing its stroke by selecting suitable parameters. Those characteristics make the DTMD a considered option for VIV control in bridges, especially when a small total mass ratio is adopted. The CBOM can be applied not only to the VIV control, but also to other vibration controls, since it has more practical application significance.

CRediT authorship contribution statement

Wen-Peng Xu: Conceptualization, Methodology, Software, Data curation, Writing – original draft, Validation, Writing – review & editing. **Jun Dai:** Conceptualization, Methodology, Software, Data curation, Writing – original draft, Validation, Writing – review & editing. **Zhao-Dong Xu:** Conceptualization, Methodology, Software, Data curation, Writing – original draft, Supervision, Validation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgements

Financial supports for this research are provided by National Natural Science Foundation of China (52108442), Changjiang Scholar Program

of Chinese Ministry of Education, Tencent Foundation through the XPLOER PRIZE, China Postdoctoral Science Foundation (2021 M690619). These supports are gratefully acknowledged.

Appendix A. . Detailed derivation of the vortex-induced amplitude and phase

The generalized response of the bridge-DTMDI system can be expressed as:

$$q_1(t) = A_1 e^{i\omega t} \# \quad (A1)$$

$$q_2(t) = A_2 e^{i\omega t} \# \quad (A2)$$

$$q_3(t) = A_3 e^{i\omega t} \# \quad (A3)$$

where A_1 , A_2 and A_3 are the complex amplitudes of the girder, TMD₁ and TMD₂. Substituting Eqs. (A.2) and (A.3) into Eqs. (5) and (6) yields:

$$\begin{aligned} \frac{A_3}{A_2} = & \frac{\left(\omega_2^2 - \frac{\beta\omega^2}{\mu_2}\right) \left[\omega_3^2 \left(1 + \frac{\mu_3}{\mu_2}\right) - \omega^2\right] + 4\xi_2\xi_3\omega_2\omega_3\omega^2 \left(1 + \frac{\mu_3}{\mu_2}\right)}{\left[\omega_3^2 \left(1 + \frac{\mu_3}{\mu_2}\right) - \omega^2\right]^2 + \left[2\xi_3\omega_3\omega \left(1 + \frac{\mu_3}{\mu_2}\right)\right]^2} \\ & + \frac{2\xi_2\omega_2\omega \left[\omega_3^2 \left(1 + \frac{\mu_3}{\mu_2}\right) - \omega^2\right] - 2\xi_3\omega_3\omega \left(\omega_2^2 - \frac{\beta\omega^2}{\mu_2}\right) \left(1 + \frac{\mu_3}{\mu_2}\right)}{\left[\omega_3^2 \left(1 + \frac{\mu_3}{\mu_2}\right) - \omega^2\right]^2 + \left[2\xi_3\omega_3\omega \left(1 + \frac{\mu_3}{\mu_2}\right)\right]^2} i \\ & = P_{32} + Q_{32}i \# \end{aligned} \quad (A4)$$

Then, $\ddot{q}_3$ can be written as:

$$\ddot{q}_3 = -\omega^2 P_{32} q_2 - \omega Q_{32} \dot{q}_2 \# \quad (A5)$$

Substituting Eq. (A.5) into Eq. (5) yields:

$$\begin{aligned} \frac{A_2}{A_1} = & \frac{\omega^2 \left(1 + \frac{\mu_3}{\mu_2}\right) \left[\omega_2^2 - \omega^2 - \frac{\beta\omega^2}{\mu_2} - \frac{\mu_3\omega^2}{\mu_2} (1 + P_{32})\right]}{\left[\omega_2^2 - \omega^2 - \frac{\beta\omega^2}{\mu_2} - \frac{\mu_3\omega^2}{\mu_2} (1 + P_{32})\right]^2 + \left(2\xi_2\omega_2\omega - \frac{\mu_3\omega^2}{\mu_2} Q_{32}\right)^2} \\ & - \frac{\omega^2 \left(1 + \frac{\mu_3}{\mu_2}\right) \left(2\xi_2\omega_2\omega - \frac{\mu_3\omega^2}{\mu_2} Q_{32}\right)}{\left[\omega_2^2 - \omega^2 - \frac{\beta\omega^2}{\mu_2} - \frac{\mu_3\omega^2}{\mu_2} (1 + P_{32})\right]^2 + \left(2\xi_2\omega_2\omega - \frac{\mu_3\omega^2}{\mu_2} Q_{32}\right)^2} i \\ & = P_{21} + Q_{21}i \# \end{aligned} \quad (A6)$$

$$\ddot{q}_2 = -\omega^2 P_{21} q_1 - \omega Q_{21} \dot{q}_1 \# \quad (A7)$$

Substituting Eq. (A.7) into Eq. (6) yields:

$$\begin{aligned} \frac{A_3}{A_1} = & \frac{(\omega_3^2 - \omega^2)\omega^2(1 + P_{21}) + 2\xi_3\omega_3\omega^3 Q_{21}}{(\omega_3^2 - \omega^2)^2 + (2\xi_3\omega_3\omega)^2} \\ & + \frac{(\omega_3^2 - \omega^2)\omega^2 Q_{21} + 2\xi_3\omega_3\omega^3(1 + P_{21})}{(\omega_3^2 - \omega^2)^2 + (2\xi_3\omega_3\omega)^2} i \\ & = P_{31} + Q_{31}i \# \end{aligned} \quad (A8)$$

$$\ddot{q}_3 = -\omega^2 P_{31} q_1 - \omega Q_{31} \dot{q}_1 \# \quad (A9)$$

Substituting Eqs. (A.7) and (A.9) into Eq. (4) yields:

$$\begin{aligned} \ddot{q}_1 + 2\xi_1\omega_1\dot{q}_1 + \omega_1^2 q_1 = & \bar{Y}_1 (\alpha_2 - \bar{\epsilon}\alpha_4 q_1^2) \dot{q}_1 + [\mu_2\omega^2(1 + P_{21}) \\ & + \mu_3\omega^2(1 + P_{21} + P_{31})] q_1 + [\mu_2\omega Q_{21} + \mu_3\omega(Q_{21} + Q_{31})] \dot{q}_1 \# \end{aligned} \quad (A10)$$

Introducing $\ddot{q}_1 + \omega_1^2 q_1 = f(q_1, \dot{q}_1)$, Eq. (A.10) can be rewritten as follows:

$$\begin{aligned} f(q_1, \dot{q}_1) = & \bar{Y}_1 (\alpha_2 - \bar{\epsilon}\alpha_4 q_1^2) \dot{q}_1 + [\mu_2\omega^2(1 + P_{21}) + \mu_3\omega^2(1 + P_{21} + P_{31})] q_1 \\ & + [\mu_2\omega Q_{21} + \mu_3\omega(Q_{21} + Q_{31}) - 2\xi_1\omega_1] \dot{q}_1 \# \end{aligned} \quad (A11)$$

Assuming that $q_1 = A_1 \cos(\omega t + \varphi)$ and $\dot{q}_1 = -\omega A_1 \sin(\omega t + \varphi)$, A_1 and φ are slow time-varying vortex-induced amplitude and phase. The following condition can be obtained:

$$\dot{A}_1 \cos(\omega t + \varphi) - A_1 \dot{\varphi} \sin(\omega t + \varphi) = 0 \# (A12).$$

Eqs. (A.11) and (A.12) yield two coupled first-order differential equations:

$$\dot{A}_1 = \frac{1}{\omega} \left\{ f(q_1, \dot{q}_1) + A_1(\omega^2 - \omega_1^2) \cos(\omega t + \varphi) \right\} \sin(\omega t + \varphi) \# (A13).$$

$$\dot{\varphi} = \frac{1}{\omega a} \left\{ f(q_1, \dot{q}_1) + A_1(\omega^2 - \omega_1^2) \cos(\omega t + \varphi) \right\} \cos(\omega t + \varphi) \# (A14).$$

According to the Krylov–Bogoliubov method [51], it is assumed that the variables A_1 and φ do not acquire substantial increments during a single period and the derivatives $\dot{A}_1$ and $\dot{\varphi}$ are constant during any one period. Therefore, the right-hand sides of Eqs. (A.13) and (A.14) can be replaced by their mean values over a period. Introducing the new variable $\tau = \omega t + \varphi$, $\dot{A}_1$ and $\dot{\varphi}$ can be expressed as:

$$\dot{A}_1 = \frac{1}{2\pi\omega} \int_0^{2\pi} \left[f(q_1, \dot{q}_1) + A_1(\omega^2 - \omega_1^2) \cos\tau \right] \sin\tau d\tau \# (A15).$$

$$\dot{\varphi} = \frac{1}{2\pi\omega a} \int_0^{2\pi} \left[f(q_1, \dot{q}_1) + A_1(\omega^2 - \omega_1^2) \cos\tau \right] \cos\tau d\tau \# (A16).$$

The $f(q_1, \dot{q}_1)$ is given by Eq. (A.11). Therefore, Eqs. (A.15) and (A.16) have the variables separate and can be integrated directly to obtain for A_1 and φ .

References

- [1] Ehsan F, Scanlan RH. Vortex-induced vibrations of flexible bridges. *J Eng Mech* 1990;116(6):1392–411. [https://doi.org/10.1061/\(ASCE\)0733-9399\(1990\)116:6\(1392\)](https://doi.org/10.1061/(ASCE)0733-9399(1990)116:6(1392)).
- [2] Williamson C, Govardhan R. A brief review of recent results in vortex-induced vibrations. *J Wind Eng Ind Aerody* 2008;96(6–7):713–35. <https://doi.org/10.1016/j.jweia.2007.06.019>.
- [3] Dai J, Xu ZD, Yin XJ, Gai PP, Luo Y. Parameters Design of TMD Mitigating Vortex-Induced Vibration of the Hong Kong-Zhuhai-Macao Bridge Deep-Water Nonnavigable Bridge. *J Bridge Eng* 2019;24(8):06019005. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0001450](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001450).
- [4] Dai J, Xu ZD, Gai PP. Parameter determination of the tuned mass damper mitigating the vortex-induced vibration in bridges. *Eng Struct* 2020;221:111084. <https://doi.org/10.1016/j.engstruct.2020.111084>.
- [5] Li H, Laima S, Zhang Q, Li N, Liu Z. Field monitoring and validation of vortex-induced vibrations of a long-span suspension bridge. *J Wind Eng Ind Aerody* 2014;124:54–67. <https://doi.org/10.1016/j.jweia.2013.11.006>.
- [6] Fujino Y, Siringoringo D. Vibration Mechanisms and Controls of Long-Span Bridges: A Review. *Struct Eng Int* 2013;23(3):248–68. <https://doi.org/10.2749/101686613X13439149156886>.
- [7] Zhang Z, Fitzgerald B. Tuned mass-damper-inerter (TMDI) for suppressing edgewise vibrations of wind turbine blades. *Eng Struct* 2020;221:110928. <https://doi.org/10.1016/j.engstruct.2020.110928>.
- [8] Xu K, Bi K, Han Q, Li X, Du X. Using tuned mass damper inerter to mitigate vortex-induced vibration of long-span bridges: Analytical study. *Eng Struct* 2019;182:101–11. <https://doi.org/10.1016/j.engstruct.2018.12.067>.
- [9] Dai J, Xu ZD, Gai PP, Xu YW. Mitigation of Vortex-Induced Vibration in Bridges Using Semiactive Tuned Mass Dampers. *J Bridge Eng* 2021;26(6):05021003. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0001719](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001719).
- [10] Martinelli L, Domaneschi M. Effect of structural control on wind fatigue mitigation in suspension bridges. *Int J Struct Eng* 2017;8(4):289–307. <https://doi.org/10.1504/IJSTRUCTE.2017.10010461>.
- [11] Ge T, Xu ZD, Guo YQ. Experimental investigation and multiscale modeling of ve damper considering chain network and ambient temperature influence. *J ENG MECH* 2022;148(1):04021124.
- [12] Dong YR, Xu ZD, Shi Q, Li QQ, He ZH, Cheng Y. Seismic performance and material-level damage evolution of retrofitted RC framed structures by high-performance AVED under different shear-span ratio. *J Building Eng* 2023;63:105495.
- [13] Dai J, Xu ZD, Gai PP, Hu ZW. Optimal design of tuned mass damper inerter with a Maxwell element for mitigating the vortex-induced vibration in bridges. *Mech Syst Signal Process* 2021;148:107180. <https://doi.org/10.1016/j.ymssp.2020.107180>.
- [14] Hu ST, Meng DL, Hu RK, Yang MG. A Combined Viscous-Steel Damping System (CVSDS) for Longitudinal Vibration Mitigation of A long-span Railway Suspension Bridge. *J Earthquake Eng* 2022.
- [15] Ge YJ, Zhao L, Cao JX. Case study of vortex-induced vibration and mitigation mechanism for a long-span suspension bridge. *J Wind Eng Ind Aerod* 2022;220. <https://doi.org/10.1016/j.jweia.2021.104866>.
- [16] Chen L, Liu ZH, Nagarajaiah S, Sun LM, Zhao L, Cui W. Vibration mitigation of long-span bridges with damped outriggers. *Eng Struct* 2022;271(15). <https://doi.org/10.1016/j.engstruct.2022.114873>.
- [17] Soong TT, Spencer BF. Supplemental energy dissipation: state-of-the-art and state-of-the-practice. *Eng Struct* 2002;24(3):243–59. [https://doi.org/10.1016/S0141-0296\(01\)00092-X](https://doi.org/10.1016/S0141-0296(01)00092-X).
- [18] Rana R, Soong TT. Parametric study and simplified design of tuned mass dampers. *Eng Struct* 1998;20(3):193–204. [https://doi.org/10.1016/S0141-0296\(97\)00078-3](https://doi.org/10.1016/S0141-0296(97)00078-3).
- [19] Marian L, Giaralis A. Optimal design of a novel tuned mass-damper-inerter (TMDI) passive vibration control configuration for stochastically support-excited structural systems. *Probab Eng Mech* 2014;38:156–64. <https://doi.org/10.1016/j.probengmech.2014.03.007>.
- [20] Zuo H, Bi K, Hao H, Ma R. Influences of ground motion parameters and structural damping on the optimum design of inerter-based tuned mass dampers. *Eng Struct* 2021;227:111422. <https://doi.org/10.1016/j.engstruct.2020.111422>.
- [21] Den Hartog JP. Mechanical vibrations. Courier Corporation; 1985.
- [22] Warburton GB. Optimum Absorber Parameters for Various Combinations of Response and Excitation Parameters. *Earthquake Eng Struct Dyn* 1982;10:381–401.
- [23] Bekdas G, Nigdeli SM. Metaheuristic Algorithms in Structural Dynamics: An Application of Tuned Mass Damper Optimization. *AIP Conf Proc* 2012;1479:1903–6. <https://doi.org/10.1063/1.4756555>.
- [24] Pourzeynali S, Salimi S, Kalesar HE. Robust multi-objective optimization design of TMD control device to reduce tall building responses against earthquake excitations using genetic algorithms. *Sci Iran* 2013;20(2):207–21. <https://doi.org/10.1016/j.scient.2012.11.015>.
- [25] Jimenez-Alonso JF, Saez A. Robust Optimum Design of Tuned Mass Dampers to Mitigate Pedestrian-Induced Vibrations Using Multi-Objective Genetic Algorithms. *Struct Eng Int* 2017;27(4):492–501. <https://doi.org/10.2749/222137917x14881937844685>.
- [26] Lavan O. Multi-objective optimal design of tuned mass dampers. *Struct Control Health Monit* 2017;24(11). <https://doi.org/10.1002/stc.2008>.
- [27] Larsena A, Svensson E, Andersen H. Design aspects of tuned mass dampers for the Great Belt East Bridge approach spans. *J Wind Eng Ind Aerody* 1995. [https://doi.org/10.1016/0167-6105\(94\)00057-K](https://doi.org/10.1016/0167-6105(94)00057-K).
- [28] Battista RC, Pfeil MS. Reduction of vortex-induced oscillations of Rio-Niterói bridge by dynamic control devices. *J Wind Eng Ind Aerody* 2000;84:273–88. [https://doi.org/10.1016/S0167-6105\(99\)00108-7](https://doi.org/10.1016/S0167-6105(99)00108-7).
- [29] Strommen E, Hjorthansen E. On the use of tuned mass dampers to suppress vortex shedding induced vibrations. *Wind struct* 2001;4(1):19–30. <https://doi.org/10.12989/was.2001.4.1.019>.
- [30] Gu M, Chen SR, Chang CC. Parametric study on multiple tuned mass dampers for buffeting control of Yangpu Bridge. *J Wind Eng Ind Aerody* 2001;89(11):987–1000. [https://doi.org/10.1016/S0167-6105\(01\)00094-0](https://doi.org/10.1016/S0167-6105(01)00094-0).
- [31] Fujino Y, Yoshida Y. Wind-induced vibration and control of Trans-Tokyo Bay Crossing bridge. *J Struct Eng* 2002;128(8):1012–25. [https://doi.org/10.1061/\(ASCE\)0733-9445\(2002\)128:8\(1012\)](https://doi.org/10.1061/(ASCE)0733-9445(2002)128:8(1012)).
- [32] Chen XH, Kareem K. Efficacy of tuned mass dampers for bridge flutter control. *J Struct Eng* 2003;129(10):1291–300. [https://doi.org/10.1061/\(ASCE\)0733-9445\(2003\)129:10\(1291\)](https://doi.org/10.1061/(ASCE)0733-9445(2003)129:10(1291)).
- [33] Domaneschi M, Martinelli L, Po E. Control of wind buffeting vibrations in a suspension bridge by TMD: Hybridization and robustness issues. *Comput Struct* 2015;155:3–17. <https://doi.org/10.1016/j.compstruc.2015.02.031>.
- [34] Abe M, Fujino Y. Dynamic Characterization of Multiple Tuned Mass Dampers and Some Design Formulas. *Earthquake Eng Struct Dyn* 1994;23(8):813–35. [https://doi.org/10.1016/0148-9062\(95\)94805-8](https://doi.org/10.1016/0148-9062(95)94805-8).
- [35] Yamaguchi H, Hampornchai N. Fundamental Characteristics of Multiple Tuned Mass Dampers for Suppressing Harmonically Forced-Oscillations. *Earthquake Eng Struct Dyn* 1993;22:51–62. <https://doi.org/10.1002/eqe.4290220105>.
- [36] Asami T. Optimal Design of Double-Mass Dynamic Vibration Absorbers Arranged in Series or in Parallel. *J Vib Acoust* 2017;139(1):011015. <https://doi.org/10.1115/1.4034776>.
- [37] Iwanami K, Seto K. An Optimum Design Method for the Dual Dynamic Damper and its Effectiveness. *Bull JSME* 1984;27(231):1965–73. <https://doi.org/10.1299/jsmes1958.27.1965>.
- [38] Xu KM, Igusa T. Dynamic Characteristics of Multiple Substructures with Closely Spaced Frequencies. *Earthquake Eng Struct Dyn* 1992;21(12):1059–70. <https://doi.org/10.1002/eqe.4290211203>.
- [39] Barredo E, Mendoza Larios JG, Mayén J, Flores-Hernández AA, Colín J, Arias MM. Optimal design for high-performance passive dynamic vibration absorbers under

- random vibration. *Eng Struct* 2019;195:469–89. <https://doi.org/10.1016/j.engstruct.2019.05.105>.
- [40] Effective ZL, Dampers R-M. *J Vib Acoust* 2009;131(3):837–88. <https://doi.org/10.1115/1.3085879>.
- [41] Ok SY, Song J, Park KS. Development of optimal design formula for bi-tuned mass dampers using multi-objective optimization. *J Sound Vib* 2009;322(1–2):60–77. <https://doi.org/10.1016/j.jsv.2008.11.023>.
- [42] Dai J, Xu ZD, Gai PP. Tuned mass-damper-inerter control of wind-induced vibration of flexible structures based on inerter location. *Eng Struct* 2019;199:109585. <https://doi.org/10.1016/j.engstruct.2019.109585>.
- [43] Smith MC. Synthesis of mechanical networks: The inerter. *IEEE Trans Autom Control* 2002;47(10):1648–62. <https://doi.org/10.1109/TAC.2002.803532>.
- [44] Ikago K, Saito K, Inoue N. Seismic control of single-degree-of-freedom structure using tuned viscous mass damper. *Earthquake Eng Struct Dyn* 2012;41(3):453–74. <https://doi.org/10.1002/eqe.1138>.
- [45] Lazar IF, Neild SA, Wagg DJ. Using an inerter-based device for structural vibration suppression. *Earthquake Eng Struct Dyn* 2014;43(8):1129–47. <https://doi.org/10.1002/eqe.2390>.
- [46] Pietrosanti D, De Angelis M, Basili M. Optimal design and performance evaluation of systems with Tuned Mass Damper Inerter (TMDI). *Earthquake Eng Struct Dyn* 2017;46(8):1367–88. <https://doi.org/10.1002/eqe.2861>.
- [47] Taflanidis AA, Giaralis A, Patsialis D. Multi-objective optimal design of inerter-based vibration absorbers for earthquake protection of multi-storey building structures. *J Franklin I* 2019;356(14):7754–84. <https://doi.org/10.1016/j.jfranklin.2019.02.022>.
- [48] Patsialis D, Taflanidis AA, Giaralis A. Tuned-mass-damper-inerter optimal design and performance assessment for multi-storey hysteretic buildings under seismic excitation. *Bull Earthquake Eng* 2021. <https://doi.org/10.1007/s10518-021-01236-4>.
- [49] Xu K, Bi KM, Ge YJ, Zhao L, Han Q, Du XL. Performance evaluation of inerter-based dampers for vortex-induced vibration control of long-span bridges: A comparative study. *Struct Control Health Monit* 2020;27(6):e2529.
- [50] Xiang HF, Chen AR, Gu M. Suppressing the vortex-excited vibration of bridges with tuned mass damper (TMD). *J Tongji Univ* 1994;22:206–11. <https://doi.org/10.1109/UIC-ATC.2009.94>.
- [51] Bogoliubov NN, Mitropolsky YA. *Asymptotic Methods in the Theory of Nonlinear Oscillations*. Gordon & Breach Science Publishers; 1961.
- [52] Qin H, Liao H, Li M, Sun Y. Vortex-Induced Vibration of Continuous Beam Bridge and Its Mitigation. Eighth Asia-Pacific Conference on Wind Engineering 2013. https://doi.org/10.3850/978-981-07-8012-8_281.
- [53] Wind-resistant Design Specification for Highway Bridges, JTG/T 3360-01-2018.
- [54] Simsek M. Non-linear vibration analysis of a functionally graded Timoshenko beam under action of a moving harmonic load. *Compos Struct* 2010;92(10):2532–46. <https://doi.org/10.1016/j.compstruct.2010.02.008>.
- [55] Dehghan-Niri E, Zahrai SM, Mohtat A. Effectiveness-robustness objectives in MTMD system design: An evolutionary optimal design methodology. *Struct Control Health Monit* 2010;17(2):218–36. <https://doi.org/10.1002/stc.297>.
- [56] Fonseca CM, Fleming PJ. An Overview of Evolutionary Algorithms in Multiobjective Optimization. *Evol Comput* 1995;3(1):1–16. <https://doi.org/10.1162/evco.1995.3.1.1>.
- [57] Deb K, Pratap A, Agarwal S, Meyarivan T. A Fast and Elitist Multiobjective Genetic Algorithm: NSGA-II. *IEEE Trans Evol Computation* 2002;6(2):182–97. <https://doi.org/10.1109/4235.996017>.
- [58] Nariman-Zadeh N, Salehpour M, Jamali A, Haghgoo E. Pareto optimization of a five-degree of freedom vehicle vibration model using a multi-objective uniform-diversity genetic algorithm (MUGA). *Eng Appl Artif Intel* 2010;23(4):543–51. <https://doi.org/10.1016/j.engappai.2009.08.008>.
- [59] Farzin H, Fotuhi-Firuzabad M, Moeini-Aghtaie M. A stochastic multi-objective framework for optimal scheduling of energy storage systems in microgrids. *IEEE Trans Smart Grid* 2016;8(1):117–27. <https://doi.org/10.1109/TSG.2016.2598678>.
- [60] Guo L, Wang N, Lu H, Li X, Wang C. Multi-objective optimal planning of the stand-alone microgrid system based on different benefit subjects. *Energy* 2016;116:353–63. <https://doi.org/10.1016/j.energy.2016.09.123>.
- [61] Coello CAC, Pulido GT, Lechuga MS. Handling multiple objectives with particle swarm optimization. *IEEE Trans evol comput* 2004;8(3):256–79. <https://doi.org/10.1109/TEVC.2004.826067>.
- [62] Marian L. The tuned mass damper inerter for passive vibration control and energy harvesting in dynamically excited structural systems. 2016.
- [63] Wang Z, Giaralis A. Enhanced motion control performance of the tuned mass damper inerter (TMDI) through primary structure shaping. *Struct Control Health Monit* 2021;28(8). <https://doi.org/10.1002/stc.2756>.